Software Requirements Specification (SRS)

Project: Mohamed Transit Group Management System

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1. Introduction

1.1. Purpose

The purpose of this document is to define the functional and non-functional requirements for the **Mohamed Transit Group Management System**. The system aims to enhance communication between the Mohamed Transit Group, importers, and internal teams, ensuring real-time updates and transparency in the importation process. The system will enable efficient tracking, live updates, and streamlined legal processes, reducing dependency on phone calls and Telegram.

1.2. Document Conventions

This document follows standard software engineering conventions, including:

- "The system shall" statements for functional requirements.
- IEEE SRS format for structure.
- UML diagrams for process representation.

1.3. Intended Audience & Reading Suggestions

The primary stakeholders for this document include:

- **Mohamed Transit Group Management:** For operational oversight and implementation.
- **Software Developers:** For designing and developing the system.
- **QA Testers:** To ensure system compliance.
- **Importers (Clients):** To understand the system's capabilities.

1.4. Scope

The Mohamed Transit Group Management System is a web-based platform designed to:

- Allow importers to track their shipments and receive live updates.
- Enable Mohamed Transit Group teams to coordinate efficiently.
- Provide real-time notifications on shipment status.
- Streamline legal document processing and approvals.
- Facilitate internal communication between teams in Mojo, Adama, and Djibouti.

2. Overall Description

2.1. Product Perspective

Currently, Mohamed Transit Group relies on phone calls and Telegram for communication, leading to inefficiencies. The proposed system will integrate live tracking, messaging, and automated notifications to modernize the workflow.

2.2. Product Functions

The system will offer the following functionalities:

1. **Importer Dashboard:** Provides live tracking, shipment status, and document updates.
2. **Internal Team Portal:** Allows team members to update shipment status and manage legal documentation.
3. **Legal Document Processing:** Digitally manages and processes importation-related legal paperwork.
4. **Communication System:** Facilitates real-time messaging between stakeholders.
5. **Role-based Access Control:** Ensures security and restricted access to sensitive data.

2.3. User Characteristics

- **Importers (Clients):** Require real-time updates, document management, and support.
- **Mohamed Transit Group Staff:** Need workflow management tools for document processing and shipment updates.

2.4. Constraints

- The system must comply with Ethiopian customs regulations.
- Real-time tracking requires stable internet connectivity.
- Data security must be GDPR & Ethiopian data protection law compliant.

2.5. Assumptions and Dependencies

- The system assumes importers have access to internet-enabled devices.
- Importers must provide valid legal representation documents.

3. Specific Requirements

3.1. Functional Requirements

3.1.1. User Authentication & Roles

- The system shall allow **importers** and **Mohamed Transit Group staff** to register and log in.
- The system shall implement **role-based access control**.
- **User roles in the system:**
  - **Manager:** Oversees system operations and assigns tasks.
  - **Assessor:** Evaluates import documents and ensures compliance.
  - **Data Encoder:** Inputs legal and shipment data into the system.
  - **Operational Agents:** Handle legal processing by physically representing the company at government offices.

3.1.2. Shipment Tracking & Live Updates

- The system shall allow importers to track their shipments in real time.
- The system shall display live shipment updates from the **Mojo, Adama, and Djibouti teams**.
- The system shall provide estimated arrival times and delivery status.

3.1.3. Legal Document Management

- The system shall allow importers to upload legal documents.
- The system shall notify staff when a document requires verification.
- The system shall store documents securely with version control.

3.1.4. Communication & Notifications

- The system shall support **real-time messaging** between importers and Mohamed Transit Group.
- The system shall send **SMS/email notifications** on shipment progress.
- The system shall generate alerts for missing or expired documents.

3.1.5. Multi-modal & Uni-modal Processing

- The system shall differentiate between multi-modal (Sea/Rail) and uni-modal (Road) operations.
- The system shall assign the Mojo team to multi-modal shipments and the Adama team to uni-modal shipments.
- The system shall track customs operations separately.

3.2. Non-Functional Requirements

3.2.1. Performance

- The system shall support **500 concurrent users** without performance degradation.
- The system shall provide updates within **2 seconds** of client changes.

3.2.2. Security

- The system shall use **strong encryption** for document storage.
- The system shall restrict access to sensitive importer data.

3.2.3. Usability & Accessibility

- The system shall be **mobile-friendly**.
- The system shall support **Amharic and English** languages.

3.2.4. Scalability

- The system shall allow future expansion to integrate new transportation hubs.

3.2.5. Compliance

- The system shall comply with **Ethiopian Data Protection Laws**.
- The system shall ensure **audit trails** for document processing.

4. Project Plan: Mohamed Transit & Logistics

The following phases will be based on iterative sprints, with clear deliverables and deadlines. Each phase includes specific tasks, meetings, and deliverables.

Phase 1: Project Kickoff & Requirement Gathering (Week 1)

Duration: 1 Week | **Start Date:** March 24 | **End Date:** March 28

Objective: Kick off the project, gather high-level requirements, and align the team with the project's goals.

Tasks: Meeting with stakeholders to define business goals, high-level understanding of functional/non-functional requirements, document initial project scope.

Deliverables: Project Scope Document, List of Key Features and Business Process Overview.

Meeting Schedule: Kickoff Meeting (March 24), Daily Standups (March 24-28).

Phase 2: Requirement Analysis & Detailed Specification (Week 2-3)

Duration: 2 Weeks | **Start Date:** March 31 | **End Date:** April 11

Objective: Gather detailed requirements, map out user stories, and produce the requirement specification document.

Tasks: Conduct stakeholder interviews, define user roles and system access, create user stories and flow diagrams.

Deliverables: User Stories & Use Case Document, System Architecture Outline, Non-Functional Requirements Document.

Meeting Schedule: Requirements Review Meeting (April 5), Daily Standups (March 31-April 10).

Phase 3: System Design & Prototyping (Week 4-5)

Duration: 3 Weeks | **Start Date:** April 14 | **End Date:** May 2

Objective: Design the system architecture, create UI/UX wireframes and prototypes, and begin front-end design.

Tasks: Design wireframes for key UI elements, create UI prototypes in Figma, design backend architecture (API and database schema).

Deliverables: Figma UI/UX Wireframes and Mockups, Front-End Codebase (Sketches), Backend Architecture Design.

Meeting Schedule: Design Review Meeting (April 25), Daily Standups (April 14 - May 2).

Phase 4: Front-End Development (Week 6-7)

Duration: 2 Weeks | **Start Date:** May 5 | **End Date:** May 23

Objective: Develop the front-end components, ensure integration with the backend, and complete initial functionality.

Tasks: Develop front-end UI components based on Figma prototypes, implement key features (forms, data display, validation), start integrating with backend APIs.

Deliverables: Functional Front-End Interface, Front-End Codebase and Documentation, Integration with Placeholder Back-End.

Meeting Schedule: Front-End Development Check-In (May 16), Daily Standups (May 5 - May 23).

Phase 5: Backend Development & Integration (Week 8-10)

Duration: 3 Weeks | **Start Date:** May 26 | **End Date:** June 13

Objective: Build the backend systems, APIs, and integrate with the front-end.

Tasks: Develop backend APIs and services for user management and data processing, set up the database, integrate front-end with backend APIs.

Deliverables: Backend APIs, Integrated System (Back-End and Front-End), Database Setup and Integration.

Meeting Schedule: Backend Review Meeting (June 9), Daily Standups (May 26 - June 13).

Phase 6: System Testing & Debugging (Week 11)

Duration: 1 Week | **Start Date:** June 16 | **End Date:** June 20

Objective: Test the integrated system, debug issues, and ensure readiness for deployment.

Tasks: Perform unit tests for front-end and back-end components, test system performance (load testing) and security, debug and finalize integration.

Deliverables: Test Results (Functional, Performance, Security), Bug Reports and Fixes Implemented, Finalized System Ready for Deployment.

Meeting Schedule: Testing Review Meeting (June 19), Daily Standups (June 16 - 20).

Phase 7: Deployment & User Acceptance Testing (UAT) (Week 12)

Duration: 1 Week | **Start Date:** June 23 | **End Date:** June 27

Objective: Deploy the system to the production environment, conduct UAT with stakeholders, and finalize project handover.

Tasks: Deploy system to production, conduct UAT with stakeholders or clients, make final system adjustments based on feedback.

Deliverables: Production Deployment (System Live), UAT Report, Final Project Sign-Off from Stakeholders.

Meeting Schedule: Deployment Review Meeting (June 25), Final Stakeholder Meeting (June 27).

Timeline Overview

Phase	Start Date	End Date	Duration	Key Deliverables
Phase 1: Kickoff & Gathering	March 24	March 28	1 Week	Project Scope, Key Features
Phase 2: Requirement Analysis	March 31	April 11	2 Weeks	User Stories, Architecture
Phase 3: System Design & Prototyping	April 14	May 2	3 Weeks	Figma Designs, Prototypes
Phase 4: Front-End Development	May 5	May 23	3 Weeks	Front-End Interface, API Integration
Phase 5: Backend Development	May 26	June 13	3 Weeks	Backend APIs, Database Setup
Phase 6: Testing & Debugging	June 16	June 20	1 Week	Test Reports, Fully Integrated System
Phase 7: Deployment & UAT	June 23	June 27	1 Week	Production Deployment, Final Sign-Off